Color theory

Concept of Color

- Color is caused by the wavelength of the light
- Humans have 3 cones in their eyes to detect 3 primary colors: red, green, blue
- There are no primary colors in nature, color spectrum is continuous
- These colors are not fixed from animal to animal: dogs see two primary colors while some insects and birds see a forth one
- The exact wavelength of each primary color changes from person to person
- The wavelength of light can be detected as a mixture of these primary colors as they overlap
- Only strange color is magenta (purple) as it is seen as a mix between blue and red, which is not possible normally
- Humans see green color brightest, blue color least bright

Color images

- Color images bring additional complexity to image processing
- While in most cases filters and operations can be performed on each channel separately, there are methods exclusive and applied differently to color images
- The most common color image format is RGB, encoding each color with one byte. This format is often extended with alpha channel, dictating opacity of each pixel when blending. There is a speed benefit of having each pixel being 32 bits apart.
- When applying convolution filters, for blur and sharpen filters, each channel should be applied separately, for edge detection you should ignore alpha channel
- In our system, ColorImage class allows you to read/write color images. Every pixel has four components: r, g, b, a.

Color models

- There are different color models apart from RGB
- RGB is popular as they could directly be used in monitors
- However, RGB is difficult to work with in many applications
- For color selection, RGB is very difficult for most people. Try to guess the color 0x804000
- RGB is also useless in physical color which is subtractive in nature
- Additionally, RGB is not physical based, summation of components do not yield to the total lightness
- H-S-L/V/I (hue, saturation, lightness, value, intensity)
  - □ Represents color using the hue of the color, saturation, and a method to represent how light is the color
  - □ HSL, HSV, HSI are different color systems
  - □ H is hue channel, it is a degree, 0 angle is red, 120 is green, 240 is blue. Yellow is at 60 degrees. Try to guess the value for orange.
  - □ S is saturation. At 0 saturation the color is a tone of gray, at 100% saturation the color is very lively
  - □ Lightness, Value, Intensity all represents the lightness of the color. All these have different calculation methods.
  - \+ Easy to convert from/to RGB
  - \+ Easier to pick/guess color
  - \- Not physical based
  - \- Even though there is a separate channel for lightness, the brightness is also affected by the hue
  - ! RGB to HSI/L/V, H has range of [0, 360), S, I, L, V are in range of [0, 1]

$$
\begin{aligned}
R' &= R/255 \\
G' &= G/255 \\
B' &= B/255 \\
C_{max} &= \max(R', G', B') \\
C_{min} &= \min(R', G', B') \\
\Delta &= C_{max} - C_{min} \\
H &= \begin{cases} 0 & , \Delta = 0 \\ 60(\frac{G'-B'}{\Delta}\ mod\ 6) & , C_{max} = R' \\ 60(\frac{B'-R'}{\Delta} + 2) & , C_{max} = G' \\ 60(\frac{R'-G'}{\Delta} + 4) & , C_{max} = B' \end{cases} \\
V &= C_{max} \\
I &= (R' + G' + B')/3 \\
L &= (C_{max} + C_{min})/2 \\
S_{HSV} &= \begin{cases} 0 & , C_{max} = 0 \\ \frac{\Delta}{C_{max}} & , otherwise \end{cases} \\
S_{HSI} &= \begin{cases} 0 & , I = 0 \\ 1 - \frac{C_{min}}{I} & , otherwise \end{cases} \\
S_{HSL} &= \begin{cases} 0 & , \Delta = 0 \\ \frac{\Delta}{1 - \|2L-1\|} & , otherwise \end{cases}
\end{aligned} \tag{1}
$$

- ! HSI/L/V to RGB

$$
\begin{aligned}
X_{HSV} &= V \times S \\
X_{HSL} &= 1 - |2L - 1| \times S \\
m_{HSV} &= V - X \\
m_{HSL} &= L - X/2 \\
Y &= X(1 - |(H/60)\ mod\ 2 - 1|) \\
R' &= m + \begin{cases} X & ,0 \leq H < 60 \\ X & ,300 \leq H < 360 \\ 0 & ,120 \leq H < 240 \\ Y & ,otherwise \end{cases} \\
G' &= m + \begin{cases} X & ,60 \leq H < 120 \\ 0 & ,240 \leq H < 300 \\ Y & ,otherwise \end{cases} \\
B' &= m + \begin{cases} 0 & ,0 \leq H < 120 \\ X & ,180 \leq H < 300 \\ Y & ,otherwise \end{cases}
\end{aligned}
\tag{2}
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- ○ CMYK
  - □ Used for printing
  - □ Subtractive (more color is darker)
  - ! RGB to CMYK. If K is 1, all other channels are 0, do not perform any calculations

$$
\begin{aligned}
R' &= R/255 \\
G' &= G/255 \\
B' &= B/255 \\
C_{max} &= \max(R', G', B') \\
K &= 1 - C_{max} \\
C &= (1 - R' - K)/(1 - K) \\
M &= (1 - G' - K)/(1 - K) \\
Y &= (1 - B' - K)/(1 - K)
\end{aligned}
\tag{3}
$$

  - ! CMYK to RGB

$$
\begin{aligned}
R &= 255 \times (1 - C) \times (1 - K) \\
G &= 255 \times (1 - M) \times (1 - K) \\
B &= 255 \times (1 - Y) \times (1 - K)
\end{aligned}
\tag{4}
$$

- ○ Lab (Lumiance, two color components) and LCh (Chromacity, hue)
  - □ These color systems are created by a consortium by averaging the color perception of many people
  - □ Lab is a regular color system while LCh is circular with hue
  - \+ L is physically based
  - \+ a, b, C, h do not have any variance in lightness
  - \+ Operations in this color space results in best results among the color systems. For instance, shifting hue channel does not affect brightness
  - \- Difficult to calculate
- ○ Grayscale
  - □ For RGB to grayscale there are different methods
  - ! L = (R + G + B) / 3
  - ! L = (max(R, G, B) + min(R, G, B)) / 2
  - ! L = 0.2126 R + 0.7152 G + 0.0722 B
  - ! R, G, B = L
- ○ There are other color system that might have special uses. For instance YUV color space is used for compression.